

Comparing a Score of 92

AP Statistics · Unit 2 · Topic 2.11 · B: 2026-11-16 · E: 2026-11-18 · TEACHER KEY

CED TETHER

- SKILLS: Using Probability and Simulation
- **Skill 3.A** — Determine relative frequencies, proportions, or probabilities using simulation or calculations.
- **Skill 3.C** — Describe probability distributions.
- ENDURING UNDERSTANDING: VAR-6 The normal distribution may be used to model variation.
- VAR-6.A Calculate the probability that a particular value lies in a given interval of a normal distribution. [Skill 3.A]
- VAR-6.A.1 A continuous random variable is a variable that can take on any value within a specified domain. Every interval within the domain has a probability associated with it. VAR-6.A.2 A continuous random variable with a normal distribution is commonly used to describe populations. The distribution of a normal random variable can be described by a normal, or "bell-shaped," curve. VAR-6.A.3 The area under a normal curve over a given interval represents the probability that a particular value lies in that interval.
- VAR-6.B Determine the interval associated with a given area in a normal distribution. [Skill 3.A]
- VAR-6.B.1 The boundaries of an interval associated with a given area in a normal distribution can be determined using z-scores or technology.
- VAR-6.C Determine the appropriateness of using the normal distribution to approximate probabilities for unknown distributions. [Skill 3.C]
- VAR-6.C.1 Normal distributions are symmetrical and "bell-shaped." As a result, normal distributions can be used to approximate distributions with similar characteristics.

Essential question: How can the same raw score have different relative standings in two normal distributions with the same mean?

DOK-3 skill: 4.B · **FRQ pattern:** same-score-two-models-compare-with-z

DOK rationale: Part (c) combines two percentile comparisons with the effect of spread to judge competing same-score claims and explain the relative standing supported by the models.

First take → rules + (a)–(c) → turn in

DOK: 1 → 3

Minutes: 45

Teacher does

- Board slide up at the bell. Ask for a claim before students calculate either percentile.
- At minute 5, read the rules box aloud once; students start (a).
- Return to the board slide at minute 33. Circulate and require both z-scores before a comparison.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose whether the percentile is the same and cite mean or standard deviation.
- Use the rules box to compare the spreads and calculate both z-scores and percentiles.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

Questions to ask

- How many School A standard deviations above 80 is a 92?

- Why does School B's wider curve place more area above the same score?
- Which area gives a percentile: left of 92 or right of 92?
- What exactly do you mean by 'better' when the raw score is identical?

Adult role

Solo teacher. Circulate 5 pairs. Require students to translate raw-point distance into standard-deviation units and define better as relative standing before choosing a school.

[WARN] LOOK FOR

- Using the shared mean as evidence that all corresponding percentiles match.
- Reporting the upper-tail probability rather than the percentile below 92.
- Reversing the comparison and assigning the higher percentile to the wider distribution.

SCORING PART (c) — E / P / I

Expected elements

- **judgment-1** (required) — Supports the data coach using percentiles about 90.9 at A and 78.8 at B.
- **judgment-2** (required) — Explains that the same 12-point gap is different numbers of standard deviations because the spreads differ.
- **judgment-3** (required) — Concludes that 92 has higher relative standing at A and explains why the shared mean is insufficient.

E	Meets all three checks with correct contextual reasoning; equivalent wording is acceptable.
P	Shows at least one substantive correct check, but has an omission or error that prevents E.
I	Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient.

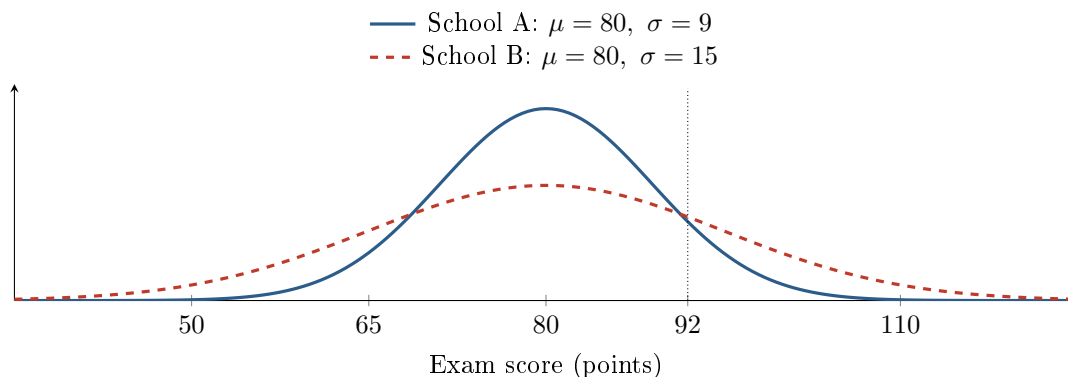
Common mistakes

- Using 80 as the standard deviation because it is shared by both models
- Computing $z = (80 - 92)/\sigma$ and reporting lower-tail areas without correcting direction
- Saying the wider distribution makes every above-mean score more exceptional
- Calling one raw score objectively higher even though the numerical score is 92 at both schools

[STAR] TODAY'S PROBLEM — annotated

Suppose exam scores at School A and School B are each approximately normal. Both distributions have mean 80 points, but School A has standard deviation 9 points and School B has standard deviation 15 points.

A counselor says, “A score of 92 is at about the 90th percentile at both schools because both schools have the same mean.” A data coach says, “The different spreads could put 92 at different percentiles.”



FIRST TAKE (5 min)

Would a 92 stand out equally at both schools? Give one reason from the picture.

Key: Not graded. Equal means should make the counselor’s claim tempting; standardization should change the final judgment.

Rules callout “STANDARDIZE BEFORE COMPARING (keep on the page)” is printed on the student sheet.

DOK 1 (a) Identify which school has the more variable score distribution and state how the corresponding normal curve shows that difference.

Key: School B is more variable because $\sigma_B = 15$ points is larger than $\sigma_A = 9$ points. Its normal curve is wider and lower; School A’s is narrower and taller around the common mean of 80.

DOK 2 (b) For a score of 92, calculate the z-score and percentile at each school. Show the standardization and report each percentile to the nearest tenth of a percent.

Key: School A: $z = (92 - 80)/9 = 1.333\dots$, so $P(X_A \leq 92) = \Phi(1.333\dots) \approx 0.9088$ and 92 is at about the 90.9th percentile. School B: $z = (92 - 80)/15 = 0.80$, so $P(X_B \leq 92) = \Phi(0.80) \approx 0.7881$ and 92 is at about the 78.8th percentile.

DOK 3 (c) In 3–4 sentences, decide whose claim is better supported. Use both percentiles from (b) and explain how the different spreads change what a score of 92 means at each school.

Key: The data coach’s claim is better supported. A 92 is about the 90.9th percentile at A and the 78.8th percentile at B. The same 12-point gap is 1.33 standard deviations at A but only 0.80 at B because the spreads differ. Thus a 92 stands higher relative to scores at School A; the shared mean alone does not settle the comparison.

EXIT

One thing the rules box changed about my first take: _____